

Experiment 20: Future Sales Prediction using Machine Learning

Aim

To implement **Future Sales Prediction** using a suitable machine learning algorithm (**Linear Regression**).

Python Program

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Sales Dataset
data = {
    'Month': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Sales': [1200, 1350, 1500, 1700, 1800,
              2000, 2200, 2400, 2600, 2800]
}

df = pd.DataFrame(data)

print("Sales Dataset:")
print(df)

# Features and Target
X = df[['Month']]
y = df['Sales']

# Create Linear Regression Model
model = LinearRegression()

# Train Model
model.fit(X, y)

# Predict Existing Data
```

```

y_pred = model.predict(X)

print("\nPredicted Sales:")
print(y_pred)

print("\nMean Squared Error:")
print(mean_squared_error(y, y_pred))

print("\nR2 Score:")
print(r2_score(y, y_pred))

# Predict Future Sales for Month 11
future_month = [[11]]
future_sales = model.predict(future_month)

print("\nPredicted Sales for Month 11:")
print(future_sales[0])

```

Sample Output

```

... Sales Dataset:
      Month  Sales
0         1   1200
1         2   1350
2         3   1500
3         4   1700
4         5   1800
5         6   2000
6         7   2200
7         8   2400
8         9   2600
9        10   2800

Predicted Sales:
[1154.54545455 1332.42424242 1510.3030303  1688.18181818 1866.06060606
 2043.93939394 2221.81818182 2399.6969697  2577.57575758 2755.45454545]

Mean Squared Error:
1187.8787878787866

R2 Score:
0.995470001762308

Predicted Sales for Month 11:
2933.333333333333

```